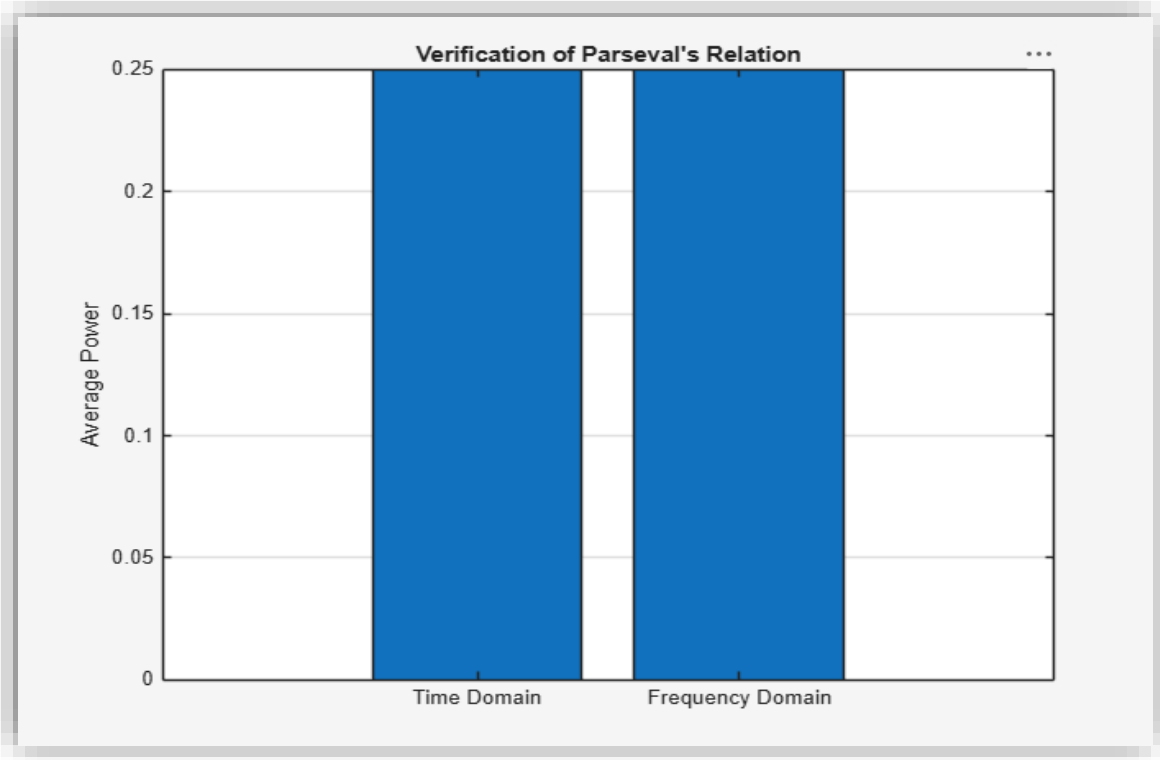
Experiment – 6
Parseval's Relation and Harmonic Spectrum Analysis

(i)

```
clc;
clear;
close all;
%% 6(i) Verification of Parseval's Relation for Periodic Signals
figure;
% Verification of Parseval's Relation
f0 = 1;
T = 1/f0;
t = linspace(0,T,4000);
x = max(sin(2*pi*f0*t),0);
% Time-domain average power
P_time = (1/T)*trapz(t,abs(x).^2);
% Fourier-series coefficients
N = 20;
a0 = (2/T)*trapz(t,x);
an = zeros(1,N); bn = zeros(1,N);
for k = 1:N
    an(k) = (2/T)*trapz(t,x.*cos(2*pi*k*f0*t));
    bn(k) = (2/T)*trapz(t,x.*sin(2*pi*k*f0*t));
end
% Frequency-domain power using Parseval's relation
P_freq = a0^2/4 + 0.5*sum(an.^2 + bn.^2);
fprintf('Time-domain power = %.6f\n',P_time);
fprintf('Frequency-domain power = %.6f\n',P_freq);
fprintf('Absolute error = %.6e\n',abs(P_time-P_freq));
bar([P_time P_freq]);
set(gca,'XTickLabel',{'Time Domain','Frequency Domain'});
ylabel('Average Power'); grid on;
title('Verification of Parseval''s Relation');
```

Code



Output